SSC MTS 2024 question paper with answers

30 September 2024 · Shift 3

79 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 11 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 15 questions

Q.1

Sharad invested two equal amounts for 6 years and 3 years at the rate of 7% and 9% per annum, respectively. The sum of their simple interest is ₹1,587. The invested amount (in ₹) by Sharad is:

(A) 4,600
(B) 4,650
(C) 4,550
(D) 4,374

Q.3

In a park there are rabbits and pigeons in the ratio 4 : 9. If legs are counted, the total number of legs is 340. Find the number of rabbits.

(A) 70
(B) 30
(C) 20
(D) 40

Q.4

By selling an article for ₹19.5, a dealer makes a profit of 10%. By how much should he increase his selling price (in ₹, rounded off to two decimal place) so as to make a profit of 50%?

(A) 8.09
(B) 10.09
(C) 7.09
(D) 9.09

Q.5

The speed of a boat in still water is 13 km/h and the speed of the stream is 4 km/h. How much more time will it take in going upstream than ingoing downstream a distance of 68 km? (Correct to the nearest minute)

(A) 3 h 33 min
(B) 3 h 30 min
(C) 2 h 55 min
(D) 3 h 50 min

Q.6

Find the simple interest (in ₹) on ₹2000 as a sum borrowed at 9% per year rate of interest for 2 years.

(A) 400
(B) 320
(C) 360
(D) 300

Q.7

In a circular race of 1500 m, P and Q start from the same point and at the same time, at speeds of 20 km/h and 35 km/h, respectively. After what time will they meet for the first time on the track if they are running in the same direction?

(A) 840 sec
(B) 640 sec
(C) 360 sec
(D) 340 sec

Q.8

Three numbers are in the ratio 3 : 4 : 5 and their HCF is 60. The numbers are:

(A) 90, 120, 150
(B) 20, 15, 12
(C) 45, 60,75
(D) 180, 240, 300

Q.10

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 9% per year rate of interest for 3 years.

(A) 850
(B) 770
(C) 750
(D) 810

Q.11

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 68 kg and 35 kg is:

(A) 48.6
(B) 51.6
(C) 50.6
(D) 49.6

Q.12

The average of the numbers 253, 245, 325, 120 and 115+2K is 212. Find the average of the numbers 353, 354, 225, 220 and 110+3K.

(A) 253
(B) 250
(C) 251
(D) 252

Q.13

The lengths of the two diagonals of a rhombus are 20 cm and 23 cm. What is the area (in cm^2) of the rhombus?

(A) 460
(B) 345
(C) 115
(D) 230

Q.14

The marked price of a chair is ₹1,800 and a discount of 18% is offered on the marked price. What additional discount must be offered to the customer now to bring the net price to ₹1,220? (Correct to two places of decimals.)

(A) 13.47%
(B) 17.34%
(C) 14.37%
(D) 31.47%

Q.15

Two candidates contested in an election. One of them secured 62.5% of the total votes and won by 996 votes. What is the total number of votes cast?(all votes are valid)

(A) 2656
(B) 2988
(C) 3984
(D) 1594

Q.18

The average weight of a class of 115 students is 52.4 kg. If the average weight of the 63 boys is 60 kg, then find the average weight (in kg) of the girls in the class. (Rounded off to one decimal place)

(A) 47.6
(B) 43.2
(C) 45.5
(D) 41.8

Q.20

The average score obtained by 64 boys and a certain number of girls in a test is 59.15, while the average score obtained by the boys is 55.25. If the average score of the girls is 68.75, then how many girls appeared for the test?

(A) 25
(B) 28
(C) 27
(D) 26

Reasoning Ability and Problem Solving 15 questions

Q.1

Which two signs should be interchanged to make the following equation mathematically correct? 49 + 7 × 10 ÷ 6 – 3 = 73

(A) + and –
(B) × and ÷
(C) – and ×
(D) + and ÷

Q.5

10 is related to 102 following a certain logic. Following the same logic, 13 is related to 171. To which of the following is 16 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying, etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 258
(B) 251
(C) 268
(D) 265

Q.6

What should come in place of the question mark (?) in the given series based on the English alphabetical order ? HAF, KDI, NGL, QJO, ?

(A) RTW
(B) SKP
(C) TMR
(D) PQR

Q.7

What will come in the place of the question mark (?) in the following equation if '+' and '–' are interchanged and '×' and '÷' are interchanged? 90 – 6 + 69 ÷ 54 × 27 = ?

(A) –44
(B) –42
(C) –43
(D) –45

Q.8

What should come in place of the question mark (?) in the given series? 162 184 206 228 250 ?

(A) 272
(B) 277
(C) 262
(D) 284

Q.9

In a certain code language, 'Do you follow' is coded as 'MP TQ IX' and 'How are you' is coded as 'TQ LV IJ'. How is 'you' coded in that code language?

(A) IX
(B) LU
(C) MP
(D) TQ

Q.11

In a certain code language, 'MICE' is coded as '6481' and 'RICE' is coded as '1584'. What is the code for 'R' in the given code language?

(A) 5
(B) 1
(C) 8
(D) 4

Q.12

S, T, U, V, W and X are sitting around a circular table facing the centre (but not necessarily in the same order). W sits second to the right of U. T sits second to the right of W. X sits to the immediate right of T. Who sits to the immediate right of X?

(A) W
(B) U
(C) S
(D) V

Q.13

Six students Dhiraj, Mohan, Lokesh, Param, Rohan and Suresh are sitting around a circular table, facing the centre (but not necessarily in the same order). Rohan is sitting second to the right of Suresh. Param is sitting third to the right of Rohan. Param is an immediate neighbour of both Lokesh and Suresh. Mohan and Lokesh are immediate neighbours. Who is the immediate neighbour of both Dhiraj and Mohan?

(A) Param
(B) Lokesh
(C) Suresh
(D) Rohan

Q.14

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: Some creepers are stones. Some stones are buds. Conclusions: (I) Some buds are stones. (II) No creeper is a bud.

(A) Both conclusions (I) and (II) follow
(B) Neither conclusion (I) nor (II) follows
(C) Only conclusion (I) follows
(D) Only conclusion (II) follows

Q.15

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (16, 3, 27) (10, 19, 37)

(A) (5, 14, 20)
(B) (15, 17, 30)
(C) (19, 15, 32)
(D) (7, 25, 40)

Q.16

UDLT is related to XGOW in a certain way based on the English alphabetical order. In the same way, ZIQY is related to CLTB. To which of the following is IQYH related, following the same logic?

(A) LTBK
(B) LTCL
(C) MKBK
(D) LGKK

Q.17

ACDB is related to ZXWY in a certain way. In the same way, EGHF is related to VTSU. To which of the following is IKLJ related, following the same logic?

(A) RPOQ
(B) RPQO
(C) PRQO
(D) PROQ

Q.18

In a certain code language, ‘clean the room’ is written as ‘rt yp fs’ and ‘room is good’ is coded as ‘sw rt la’. How is ‘room’ coded in the given language

(A) yp
(B) sw
(C) rt
(D) fs

Q.20

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some actors are dancers. Some dancers are singers. Conclusions: (I) Some actors are singers. (II) No singer is a dancer.

(A) Only conclusion II follows
(B) Neither conclusion I nor II follows
(C) Only conclusion I follows
(D) Both conclusions I and II follow

General Awareness 25 questions

Q.1

On boiling an egg, why does the outer part become opaque and white?

(A) The heat causes a change in the crystal structure of egg yolk.

(B) The heat causes a change in the structure of egg protein.

(C) The heat causes denaturation, hardening the albumin and colouring it opaque white

(D) The heat causes a reaction of the egg with water and makes it opaque.

Q.2

The first post-independence National Games were hosted by which of the following cities?

(A) Lucknow

(B) Madras

(C) Jabalpur

(D) Delhi

Q.3

Which of the following National Parks is located in Madhya Pradesh?

(A) Periyar

(B) Silent Valley

(C) Nanda Devi

(D) Bandhavgarh

Q.4

R Muthukannammal, who was awarded the Padma Shri in 2022, is a _____ dancer.

(A) Rouf

(B) Garba

(C) Chhau

(D) Sadir

Q.5

Calculate the Body Mass Index (BMI) of a person whose weight is 50 kg and height is 1.5 m.

(A) 22.2 kg/m^2

(B) 25.0 kg/m^2

(C) 18.0 kg/m^2

(D) 16.6 kg/m^2

Q.6

On 27th october 2019, Dushyant Chautala took an oath as Deputy Chief Minister of which of the following states?

(A) Punjab

(B) Madhya Pradesh

(C) Himachal Pradesh

(D) Haryana

Q.7

Which of the following states of India marked the lowest in National Multi-dimensional Poverty Index: A Progress Review 2023?

(A) Sikkim

(B) Kerala

(C) Tamil Nadu

(D) Goa

Q.8

Where was the Hindu college established in the year 1791 to teach ancient Sanskrit texts?

(A) Calcutta

(B) Madras

(C) Agra

(D) Banaras

Q.9

Identify the INCORRECT pair of cropping season and related crop from the following.

(A) Kharif - Maize

(B) Kharif – Soyabean

(C) Rabi - Wheat

(D) Rabi - Paddy

Q.10
The Ganesh-rath is one of the finest monolithic temples at _____.

(A) Sarnath
(B) Mahabalipuram
(C) Gangaikondacholapuram
(D) Kanchipuram

Q.11
The average separation between the sun and the earth in terms of light years is 8.311 minutes. Here, minutes indicate:

(A) distance
(B) time
(C) velocity
(D) speed

Q.12
The famous Hindi novel 'Maila Aanchal' is written by:

(A) Premchand
(B) Shrilal Shukla
(C) Phanishwar Nath Renu
(D) Mridula Garg

Q.13
Which of the following is an Indian folk dance?

(A) Mohiniyattam
(B) Bihu
(C) Bharatanatyam
(D) Sattriya

Q.14
Which of the following is NOT a part of the capital receipts?

(A) Borrowing
(B) Disinvestment
(C) Tax revenue
(D) Recovery of loan

Q.15
Which of the following is the compound with the chemical formula H_2SO_4?

(A) Sulfurous Acid
(B) Sulfuric Acid
(C) Hydrogen Sulfide
(D) Nitric Acid

Q.16
According to Article 74(1), who is at the head of the Council of Ministers?

(A) Speaker of the House
(B) President
(C) Prime Minister
(D) Chief Justice

Q.17
The officer known as Diwan belonged to which of the following departments of the Mughal empire during Akbar's reign?

(A) Finance
(B) Judicial
(C) Foreign affairs
(D) Military

Q.18
Who among the following personalities was the recipient of the Lata Mangeshkar Award for Music 2021, presented by the Madhya Pradesh government?

(A) Darshan Raval
(B) Shaan
(C) Udit Narayan
(D) Kumar Sanu

Q.19
In which of the following states is the Behdienkhlam Festival celebrated, in which the young men make a symbolic gesture of driving away of the evil spirit, plague and disease by beating the roof of every house with bamboo poles?

(A) Meghalaya
(B) Odisha
(C) Manipur
(D) Bihar

Q.20
Jyotiba Phule belonged from which state?

(A) Gujarat
(B) Karnataka
(C) Tamil Nadu
(D) Maharashtra

Q.21
The word ‘Megalith’ comes from two _____ words; ‘megas’ means great and ‘lithas’ means stone.

(A) Greek
(B) English
(C) Latin
(D) Hebrew

Q.22
Which of the following Articles on Fundamental Duties was added to the Indian Constitution by the 42nd Constitutional Amendment Act?

(A) 52-A
(B) 45A
(C) 51-A
(D) 40

Q.23
Which country won the 4th Asian Kho-Kho Championship 2023 event in the Men's category?

(A) Iran
(B) India
(C) South Korea
(D) Indonesia

Q.24
The Hindu temples ‘Hinglaj Mata’ and ‘Katasraj’ are located in which neighbouring country of India?

(A) Sri Lanka
(B) Bhutan
(C) Nepal
(D) Pakistan

Q.25
Where was Global Technology Summit 2022 held?

(A) Bengaluru
(B) Jaipur
(C) Varanasi
(D) New Delhi

English Language and Comprehension 24 questions

Q.1
Select the most appropriate ANTONYM of the underlined word. Rakesh committed the mistake, and he must be “convicted”.

(A) Acquitted
(B) Punished
(C) Crucified
(D) Sentenced

Q.2
Select the most appropriate synonym of the underlined word in the given sentence. A “detrimental” attitude can always be cured with a cordial approach.

(A) unmindful
(B) cooperative
(C) damaging
(D) social

Q.3

Select the most appropriate option that can substitute the underlined segment in the given sentence. It is essential that you “act in accordance” with the rules and regulations of the organisation.

(A) comply
(B) observe
(C) obey
(D) fulfill

Q.4

Select the most appropriate option to fill in the blank. He needs to _____ as his exams are approaching.

(A) bury himself in
(B) learn the ropes
(C) hit the sack
(D) hit the books

Q.5

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don’t find any error, mark ‘No error’ as your answer. By next December / we will has lived / in this city for 28 years.

(A) in this city for 28 years
(B) we will has lived
(C) by next December
(D) no error

Q.6

Select the most appropriate synonym of the underlined word in the given sentence. It is impossible to think of Lady Stranleigh as an “ordinary” mother.

(A) Unlucky
(B) Blessed
(C) Usual
(D) Strange

Q.7

Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select ‘No substitution’. “If you do not followed the rules”, you will not be allowed to stay at this hostel.

(A) If you should not follow the rules
(B) If you do not follow the rules
(C) No substitution
(D) If you cannot be following the rules

Q.8

The following sentence has been split into four segments. Identify the segment that contains an error. A Gita is considered / to be a dialogue between / the Pandava prince, Arjuna, / and Lord Krishna.

(A) A Gita is considered
(B) and Lord Krishna
(C) to be a dialogue between
(D) the Pandava prince, Arjuna

Q.9

Select the most appropriate word to replace the underlined word in the following sentence. The officer was “polite” to the workers.

(A) officious
(B) harsh
(C) humble
(D) adamant

Q.10

Which of the following sentences contains an error in the use of a phrasal verb?

(A) Can you please take care for my belongings for some time?
(B) While visiting the library, I came across an interesting book.
(C) Let's call off the picnic as there is a possibility of heavy rain.
(D) It is wise to put up with one's quarrelsome neighbours.

Q.11

The given sentence is divided into four segments. Select the option that has the segment with the grammatical error. Mahesh and his workers / will have been worked / on the project / for a long time.

(A) for a long time
(B) Mahesh and his workers
(C) on the project
(D) will have been worked

Q.12

Select the correct spelling of the underlined word. "Today's “whether” is very pleasant", said Maya.

(A) Wether
(B) Weather
(C) Weathur
(D) Wheather

Q.13

Select the INCORRECTLY spelt word.

(A) Cemetery
(B) Accommodate
(C) Calender
(D) Conscience

Q.14

Select the most appropriate option that can substitute the underlined word in the given sentence. Mita is “smart” than many other kids in the school.

(A) most smart
(B) smartest
(C) smarter
(D) less smart

Q.16

Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select ‘No substitution’. “After she will get her degree”, she moved to Germany.

(A) No substitution
(B) After she got her degree
(C) After she gets her degree
(D) After she get her degree

Q.17

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. In accordance to / the factual data, / serious measures are required / to control the loss.

(A) serious measures are required
(B) In accordance to
(C) the factual data
(D) to control the loss

Q.18

Select the most appropriate option to fill in the blank and complete the phrase/idiom. When Jane realised that she had made a serious mistake at work, she knew she had to face the _____and take responsibility for her actions.

(A) mirror
(B) fun
(C) music
(D) band

Q.19

Select the most appropriate option to fill in the blank. The billionaire knows how to _____ money.

(A) develop
(B) do
(C) form
(D) make

Q.20

Select the most appropriate synonym of the underlined word. The “lavish” party had a luxurious atmosphere and elegant decorations.

(A) Excessive
(B) Middling
(C) Simple
(D) Happy

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Credit cards originated in the 1920s in the US, with European references dating back to 1890. John Biggins, 1._____ of the first bank-issued credit card, invented the 'Charge-It' programme in 1946, 2._____ bank customers and local merchants. In 1950, Diners Club introduced their credit card in the US, 3._____ customers to pay restaurant bills without cash. American Express 4._____ their first credit card in 1958, while the Bank of America introduced the Bank America Card (now Visa) in 1958. Visa, now known as the Visa International Service Association, is accepted in over 150 countries, making it one of the most 5._____ credit cards available.

Select the most appropriate option to fill in blank number 1.

(A) discoverer
(B) inventor
(C) finder
(D) producer

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Credit cards originated in the 1920s in the US, with European references dating back to 1890. John Biggins, 1._____ of the first bank-issued credit card, invented the 'Charge-It' programme in 1946, 2._____ bank customers and local merchants. In 1950, Diners Club introduced their credit card in the US, 3._____ customers to pay restaurant bills without cash. American Express 4._____ their first credit card in 1958, while the Bank of America introduced the Bank America Card (now Visa) in 1958. Visa, now known as the Visa International Service Association, is accepted in over 150 countries, making it one of the most 5._____ credit cards available.

Select the most appropriate option to fill in blank number 2.

(A) connecting
(B) smashing
(C) untangling
(D) splitting

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Credit cards originated in the 1920s in the US, with European references dating back to 1890. John Biggins, 1._____ of the first bank-issued credit card, invented the 'Charge-It' programme in 1946, 2._____ bank customers and local merchants. In 1950, Diners Club introduced their credit card in the US, 3._____ customers to pay restaurant bills without cash. American Express 4._____ their first credit card in 1958, while the Bank of America introduced the Bank America Card (now Visa) in 1958. Visa, now known as the Visa International Service Association, is accepted in over 150 countries, making it one of the most 5._____ credit cards available.

Select the most appropriate option to fill in blank number 3.

(A) enjoying
(B) prohibiting
(C) forbidding
(D) allowing

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Credit cards originated in the 1920s in the US, with European references dating back to 1890. John Biggins, 1._____ of the first bank-issued credit card, invented the 'Charge-It' programme in 1946, 2._____ bank customers and local merchants. In 1950, Diners Club introduced their credit card in the US, 3._____ customers to pay restaurant bills without cash. American Express 4._____ their first credit card in 1958, while the Bank of America introduced the Bank America Card (now Visa) in 1958. Visa, now known as the Visa International Service Association, is accepted in over 150 countries, making it one of the most 5._____ credit cards available.

Select the most appropriate option to fill in blank number 4.

(A) issued
(B) censored
(C) published
(D) suppressed

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Credit cards originated in the 1920s in the US, with European references dating back to 1890. John Biggins, 1._____ of the first bank-issued credit card, invented the 'Charge-It' programme in 1946, 2._____ bank customers and local merchants. In 1950, Diners Club introduced their credit card in the US, 3._____ customers to pay restaurant bills without cash. American Express 4._____ their first credit card in 1958, while the Bank of America introduced the Bank America Card (now Visa) in 1958. Visa, now known as the Visa International Service Association, is accepted in over 150 countries, making it one of the most 5._____ credit cards available.

Select the most appropriate option to fill in blank number 5.

(A) entitle
(B) solidify
(C) universal
(D) amateur

Answer key

Numerical and Mathematical Ability

Q.1 **(A)**	Q.3 **(D)**	Q.4 **(C)**	Q.5 **(A)**	Q.6 **(C)**	Q.7 **(C)**
Q.8 **(D)**	Q.10 **(D)**	Q.11 **(D)**	Q.12 **(A)**	Q.13 **(D)**	Q.14 **(B)**
Q.15 **(C)**	Q.18 **(B)**	Q.20 **(D)**			

Reasoning Ability and Problem Solving

Q.1 **(D)**	Q.5 **(A)**	Q.6 **(C)**	Q.7 **(B)**	Q.8 **(A)**	Q.9 **(D)**
Q.11 **(A)**	Q.12 **(B)**	Q.13 **(D)**	Q.14 **(C)**	Q.15 **(D)**	Q.16 **(A)**
Q.17 **(A)**	Q.18 **(C)**	Q.20 **(B)**			

General Awareness

Q.1 **(C)**	Q.2 **(A)**	Q.3 **(D)**	Q.4 **(D)**	Q.5 **(A)**	Q.6 **(D)**
Q.7 **(B)**	Q.8 **(D)**	Q.9 **(D)**	Q.10 **(B)**	Q.11 **(A)**	Q.12 **(C)**
Q.13 **(B)**	Q.14 **(C)**	Q.15 **(B)**	Q.16 **(C)**	Q.17 **(A)**	Q.18 **(D)**
Q.19 **(A)**	Q.20 **(D)**	Q.21 **(A)**	Q.22 **(C)**	Q.23 **(B)**	Q.24 **(D)**
Q.25 **(D)**					

English Language and Comprehension

Q.1 **(A)**	Q.2 **(C)**	Q.3 **(A)**	Q.4 **(D)**	Q.5 **(B)**	Q.6 **(C)**
Q.7 **(B)**	Q.8 **(A)**	Q.9 **(C)**	Q.10 **(A)**	Q.11 **(D)**	Q.12 **(B)**
Q.13 **(C)**	Q.14 **(C)**	Q.16 **(B)**	Q.17 **(B)**	Q.18 **(C)**	Q.19 **(D)**
Q.20 **(A)**	Q.21 **(B)**	Q.22 **(A)**	Q.23 **(D)**	Q.24 **(A)**	Q.25 **(C)**

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (A) 4,600

With P in each, the two interests total P × 7 × 6/100 + P × 9 × 3/100 = 0.69P, so 0.69P = 1,587 and P = ₹2,300. The two equal amounts total ₹4,600.

Q.3 · Answer (D) 40

Rabbits have 4 legs and pigeons 2: 4 × 4x + 2 × 9x = 34x = 340, so x = 10 and there are 4 × 10 = 40 rabbits.

Q.4 · Answer (C) 7.09

CP = 19.5/1.1 = ₹17.727. For a 50% profit the price must be 1.5 × 17.727 = ₹26.59, an increase of 26.59 − 19.5 = ₹7.09.

Q.5 · Answer (A) 3 h 33 min

Upstream the boat makes 13 − 4 = 9 km/h and downstream 13 + 4 = 17 km/h. 68/9 − 68/17 = 7.556 − 4 = 3.556 hours, about 3 h 33 min.

Q.6 · Answer (C) 360

SI = 2,000 × 9 × 2/100 = ₹360.

Q.7 · Answer (C) 360 sec

Running the same way, Q gains 35 − 20 = 15 km/h = 25/6 m/s on P. They meet when Q has gained one full lap: 1,500 × 6/25 = 360 seconds.

Q.8 · Answer (D) 180, 240, 300

The numbers are 3k, 4k and 5k, whose HCF is k (3, 4 and 5 have no common factor). With k = 60 they are 180, 240 and 300.

Q.10 · Answer (D) 810

SI = 3,000 × 9 × 3/100 = ₹810.

Q.11 · Answer (D) 49.6

(40 + 49 + 56 + 68 + 35)/5 = 248/5 = 49.6 kg.

Q.12 · Answer (A) 253

The first five total 5 × 212 = 1,060 and 253 + 245 + 325 + 120 = 943, so 115 + 2K = 117 and K = 1. The second set is 353 + 354 + 225 + 220 + 113 = 1,265: average 1,265/5 = 253.

Q.13 · Answer (D) 230

Area of a rhombus = ½ × d_1 × d_2 = ½ × 20 × 23 = 230 cm².

Q.14 · Answer (B) 17.34%

After 18% off the price is 0.82 × 1,800 = ₹1,476. To bring it to ₹1,220 the further discount is 256/1,476 × 100 = 17.34%.

Q.15 · Answer (C) 3984

The winner had 62.5% and the loser 37.5%, so the margin is 25% of the votes: 996 ÷ 0.25 = 3,984.

Q.18 · Answer (B) 43.2

All 115 students weigh 115 × 52.4 = 6,026 kg and the 63 boys 63 × 60 = 3,780 kg. The 52 girls weigh 2,246 kg: 2,246/52 = 43.2 kg.

Q.20 · Answer (D) 26

With g girls: 64(59.15 − 55.25) = g(68.75 − 59.15), so 249.6 = 9.6g and g = 26.

Reasoning Ability and Problem Solving

Q.1 · Answer (D) + and ÷

Interchanging + and ÷: 49 ÷ 7 × 10 + 6 − 3 = 70 + 6 − 3 = 73.

Q.5 · Answer (A) 258
The second number is the square of the first plus 2: $10^2 + 2 = 102$ and $13^2 + 2 = 171$. Likewise $16^2 + 2 = 258$.

Q.6 · Answer (C) TMR
Every letter moves 3 places forward each time: H, K, N, Q, T; A, D, G, J, M; F, I, L, O, R. The next cluster is TMR.

Q.7 · Answer (B) –42
After interchanging the signs: $90 + 6 - 69 \times 54 \div 27 = 96 - 138 = -42$.

Q.8 · Answer (A) 272
Each term is 22 more than the one before: 250 + 22 = 272.

Q.9 · Answer (D) TQ
'you' is the only word in both sentences and 'TQ' the only code in both, so you = TQ.

Q.11 · Answer (A) 5
I, C and E are in both words, as are 1, 4 and 8 in both codes. The letter left in RICE is R and the digit left in its code is 5, so R = 5.

Q.12 · Answer (B) U
Numbering the seats to the right from U (0): W is at 2, T at 4 and X at 5. The next seat to the right of X is U's.

Q.13 · Answer (D) Rohan
Numbering the seats to the right from Suresh (0): Rohan is at 2 and Param at 5, beside Suresh. Lokesh, Param's other neighbour, is at 4, Mohan beside him at 3, and Dhiraj at 1. Rohan, at 2, sits between Dhiraj and Mohan.

Q.14 · Answer (C) Only conclusion (I) follows
'Some stones are buds' converts to 'some buds are stones', so I follows. Nothing links creepers to buds, so II does not.

Q.15 · Answer (D) (7, 25, 40)
The third number is the sum of the first two plus 8: 16 + 3 + 8 = 27 and 10 + 19 + 8 = 37. Likewise 7 + 25 + 8 = 40.

Q.16 · Answer (A) LTBK
Each letter moves 3 places forward (Y wraps round to B): UDLT → XGOW and ZIQY → CLTB. So IQYH → LTBK.

Q.17 · Answer (A) RPOQ
Each letter is replaced by its opposite in the alphabet (A ↔ Z, B ↔ Y and so on): ACDB → ZXWY and EGHF → VTSU. So IKLJ → RPOQ.

Q.18 · Answer (C) rt
'room' is the only word in both sentences and 'rt' the only code in both, so room = rt.

Q.20 · Answer (B) Neither conclusion I nor II follows
Actors and singers are each linked only to some dancers, so I need not follow. II contradicts 'some dancers are singers'. Neither follows.

General Awareness

Q.1 · Answer (C) The heat causes denaturation, hardening the albumin and colouring it opaque white
SSC's key gives the fuller answer: heat denatures albumin, the egg-white protein, whose unfolded molecules bind into a firm mesh that scatters light, so the clear white turns opaque white. (The statement that heat changes the structure of egg protein describes the same process less fully.)

Q.2 · Answer (A) Lucknow
The first National Games after independence were held at Lucknow in 1948.

Q.3 · Answer (D) Bandhavgarh
Bandhavgarh National Park is in Umaria district, Madhya Pradesh. Periyar and Silent Valley are in Kerala, Nanda Devi in Uttarakhand.

Q.4 · Answer (D) Sadir
R. Muthukannammal of Viralimalai, Tamil Nadu, received the Padma Shri in 2022 for Sadir, the temple dance tradition from which Bharatanatyam developed.

Q.5 · Answer (A) 22.2 kg/m²
BMI = weight ÷ height² = 50 ÷ 1.5^2 = 50 ÷ 2.25 = 22.2 kg/m².

Q.6 · Answer (D) Haryana
Dushyant Chautala of the Jannayak Janta Party was sworn in as Deputy Chief Minister of Haryana on 27 October 2019, alongside Chief Minister Manohar Lal Khattar.

Q.7 · Answer (B) Kerala
In NITI Aayog's National Multidimensional Poverty Index: A Progress Review 2023, Kerala had the lowest share of multidimensionally poor people (0.55%), ahead of Goa (0.84%), Tamil Nadu (2.20%) and Sikkim (2.60%).

Q.8 · Answer (D) Banaras
Jonathan Duncan, the East India Company's Resident, founded the Sanskrit College at Banaras (Varanasi) in 1791 for the study of Hindu law and philosophy.

Q.9 · Answer (D) Rabi - Paddy
Paddy (rice) is mainly a kharif crop, sown with the monsoon. Wheat is the main rabi crop, and maize and soyabean are kharif crops.

Q.10 · Answer (B) Mahabalipuram
The Ganesha Ratha is one of the Pallava monolithic rock-cut temples at Mahabalipuram (Mamallapuram), Tamil Nadu.

Q.11 · Answer (A) distance
SSC's key gives distance: like the light year, the light minute is a unit of distance, the distance light travels in one minute (about 1.8 crore km), so 8.311 light minutes measures how far the Sun is. (The figure is also the time light takes to reach the Earth, which makes 'time' tempting.)

Q.12 · Answer (C) Phanishwar Nath Renu
'Maila Aanchal' (1954), a landmark of the regional (aanchalik) Hindi novel, was written by Phanishwar Nath Renu.

Q.13 · Answer (B) Bihu
Bihu is the folk dance of Assam. Mohiniyattam, Bharatanatyam and Sattriya are among India's classical dances.

Q.14 · Answer (C) Tax revenue
Tax revenue is a revenue receipt. Borrowing, disinvestment and the recovery of loans either create a liability or reduce an asset, so they are capital receipts.

Q.15 · Answer (B) Sulfuric Acid
H_2SO_4 is sulfuric acid. Sulfurous acid is H_2SO_3, hydrogen sulfide H_2S and nitric acid HNO_3.

Q.16 · Answer (C) Prime Minister
Article 74(1) provides a Council of Ministers with the Prime Minister at its head to aid and advise the President.

Q.17 · Answer (A) Finance
Under Akbar the diwan (diwan-i-kul) was in charge of revenue and finance.

Q.18 · Answer (D) Kumar Sanu
Madhya Pradesh's National Lata Mangeshkar Award for 2021 went to the playback singer Kumar Sanu; it was presented in Indore in September 2022.

Q.19 · Answer (A) Meghalaya
Behdienkhlam is celebrated by the Pnar (Jaintia) people, above all at Jowai in Meghalaya's Jaintia Hills.

Q.20 · Answer (D) Maharashtra
Jyotirao (Jyotiba) Phule, the social reformer who founded the Satyashodhak Samaj in Pune in 1873, was from Maharashtra.

Q.21 · Answer (A) Greek
The word is Greek: 'megas', great, and 'lithos', stone. Megaliths are large stones, often set up to mark burial sites.

Q.22 · Answer (C) 51-A
Article 51A (Part IVA), added by the 42nd Amendment in 1976, lists the Fundamental Duties.

Q.23 · Answer (B) India
India won the men's title at the 4th Asian Kho-Kho Championship at Tamulpur, Assam, in March 2023, beating Nepal in the final; the Indian women also won.

Q.24 · Answer (D) Pakistan
Hinglaj Mata is in Balochistan and the Katas Raj temples are in Punjab, both in Pakistan.

Q.25 · Answer (D) New Delhi
The 2022 Global Technology Summit, hosted by Carnegie India with the Ministry of External Affairs, was held in New Delhi from 29 November to 1 December 2022.

English Language and Comprehension

Q.1 · Answer (A) Acquitted
To be 'convicted' is to be found guilty; its opposite is 'acquitted', found not guilty.

Q.2 · Answer (C) damaging
'Detrimental' means causing harm: 'damaging'.

Q.3 · Answer (A) comply
'Comply with' means act in accordance with. 'Observe', 'obey' and 'fulfill' do not take the 'with' that follows.

Q.4 · Answer (D) hit the books
To 'hit the books' is to study hard. To 'hit the sack' is to go to bed and to 'learn the ropes' is to learn how a job is done; 'bury himself in' needs an object.

Q.5 · Answer (B) we will has lived
After 'will' the verb takes its base form: 'we will have lived' (future perfect).

Q.6 · Answer (C) Usual
'Ordinary' means usual or commonplace.

Q.7 · Answer (B) If you do not follow the rules
After 'do not' the verb takes its base form: 'If you do not follow the rules'.

Q.8 · Answer (A) A Gita is considered
A particular, well-known text takes the definite article: 'The Gita is considered to be a dialogue ...'.

Q.9 · Answer (C) humble
'Humble' (modest and respectful in manner) is the closest of the options to 'polite'; 'officious' means bossily interfering, and 'harsh' and 'adamant' do not fit.

Q.10 · Answer (A) Can you please take care for my belongings for some time?
The phrasal verb is 'take care of', not 'take care for'. 'Came across', 'call off' and 'put up with' are used correctly.

Q.11 · Answer (D) will have been worked
Mahesh and his workers are doing the work, so the active future perfect continuous is needed: 'will have been working'.

Q.12 · Answer (B) Weather
'Weather' is the state of the atmosphere; 'whether' introduces alternatives.

Q.13 · Answer (C) Calender
The correct spelling is 'calendar'.

Q.14 · Answer (C) smarter
With 'than' the comparative is needed: 'smarter than many other kids'.

Q.16 · Answer (B) After she got her degree
The main clause is in the past ('moved'), so the time clause takes the past too: 'After she got her degree'.

Q.17 · Answer (B) In accordance to
The phrase is 'in accordance with': 'In accordance with the factual data'.

Q.18 · Answer (C) music
To 'face the music' is to accept the consequences of one's actions.

Q.19 · Answer (D) make
Money is something one 'makes' (earns); one does not 'do', 'form' or 'develop' money.

Q.20 · Answer (A) Excessive
'Lavish' means extravagant, spent or provided in great, even excessive, quantity; of the options only 'excessive' is close. 'Middling' and 'simple' are nearer its opposite.

Q.21 · Answer (B) inventor
Biggins 'invented' the Charge-It programme, so he was the 'inventor' of the first bank-issued card.

Q.22 · Answer (A) connecting
The programme linked bank customers with local merchants: 'connecting' them.

Q.23 · Answer (D) allowing
The card let customers pay restaurant bills without cash: 'allowing' them to.

Q.24 · Answer (A) issued
A card company 'issues' a credit card: American Express 'issued' its first in 1958.

Q.25 · Answer (C) universal
Accepted in over 150 countries, Visa is one of the most 'universal' (widely accepted) cards.

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